

world-famous-place-eda-1

January 11, 2026

1 Project Name: World Famous Places Analysis

1.0.1 Project Overview

This project explores a dataset of world-famous tourist places to understand global tourism patterns.

The dataset includes information such as annual visitors, entry fees, UNESCO status, region, and type of attraction.

The analysis focuses on identifying popular destinations and understanding factors that influence tourist visits

1.0.2 Objectives

- To explore and understand global tourism trends using data from world-famous tourist destinations
- To analyze how visitor numbers vary across countries, cities, and regions
- To examine the impact of UNESCO World Heritage status on tourist visits
- To study the relationship between entry fees and annual visitors
- To identify the most popular types of tourist attractions
- To derive meaningful insights that explain what makes a tourist place highly visited

```
[ ]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
```

```
[5]: #Load Dataset
df = pd.read_csv("world_famous_places_2024.csv")
```

Dataset Preview

```
[ ]: #First 5 rows
df.head()
```

The dataset contains information about 30 world-famous tourist places across different countries, cities, and regions. Each record includes visitor numbers, entry fee, UNESCO status, and tourism-related attributes.

Dataset Structure

```
[7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 30 entries, 0 to 29
```

```
Data columns (total 13 columns):
```

#	Column	Non-Null Count	Dtype
0	Place_Name	30 non-null	object
1	Country	30 non-null	object
2	City	30 non-null	object
3	Annual_Visitors_Millions	30 non-null	float64
4	Type	30 non-null	object
5	UNESCO_World_Heritage	30 non-null	object
6	Year_Built	30 non-null	object
7	Entry_Fee_USD	30 non-null	int64
8	Best_Visit_Month	30 non-null	object
9	Region	30 non-null	object
10	Tourism_Revenue_Million_USD	30 non-null	int64
11	Average_Visit_Duration_Hours	30 non-null	float64
12	Famous_For	30 non-null	object

```
dtypes: float64(2), int64(2), object(9)
```

```
memory usage: 3.2+ KB
```

The dataset has 13 columns. All columns have 30 non-null values, indicating a complete and well-structured dataset.

```
[ ]:
```

```
[8]: df.describe()
```

```
[8]:
```

	Annual_Visitors_Millions	Entry_Fee_USD	Tourism_Revenue_Million_USD	\
count	30.000000	30.000000	30.000000	
mean	11.331667	24.600000	593.500000	
std	12.044247	28.572473	1521.481894	
min	1.500000	0.000000	25.000000	
25%	4.775000	0.000000	61.250000	
50%	7.575000	19.000000	87.500000	
75%	12.250000	35.000000	167.500000	

max	50.000000	109.000000	6800.000000
-----	-----------	------------	-------------

	Average_Visit_Duration_Hours
count	30.000000
mean	3.166667
std	2.499425
min	0.500000
25%	1.500000
50%	2.250000
75%	4.000000
max	10.000000

Missing Values Check

```
[9]: df.isnull().sum()
```

```
[9]: Place_Name      0
     Country        0
     City           0
     Annual_Visitors_Millions  0
     Type           0
     UNESCO_World_Heritage    0
     Year_Built        0
     Entry_Fee_USD        0
     Best_Visit_Month        0
     Region           0
     Tourism_Revenue_Million_USD  0
     Average_Visit_Duration_Hours  0
     Famous_For        0
     dtype: int64
```

There are no missing values in the dataset, so no data cleaning or imputation is required before analysis.

```
[10]: print("First 5 rows:\n", df.head())
      print("\nData Info:\n")
      df.info()
      print("\nMissing Values:\n", df.isnull().sum())
      print("\nSummary Statistics:\n", df.describe(include='all'))
```

First 5 rows:

	Place_Name	Country	City \
0	Eiffel Tower	France	Paris
1	Times Square	United States	New York City
2	Louvre Museum	France	Paris
3	Great Wall of China	China	Beijing/Multiple
4	Taj Mahal	India	Agra

	Annual_Visitors_Millions	Type	UNESCO_World_Heritage	\
0	7.0	Monument/Tower	No	
1	50.0	Urban Landmark	No	
2	8.7	Museum	Yes	
3	10.0	Historic Monument	Yes	
4	7.5	Monument/Mausoleum	Yes	

	Year_Built	Entry_Fee_USD	Best_Visit_Month	Region	\
0	1889	35	May-June/Sept-Oct	Western Europe	
1	1904	0	Apr-June/Sept-Nov	North America	
2	1793	22	Oct-March	Western Europe	
3	220 BC - 1644 AD	10	Apr-May/Sept-Oct	East Asia	
4	1653	15	Oct-March	South Asia	

	Tourism_Revenue_Million_USD	Average_Visit_Duration_Hours	\
0	95	2.5	
1	70	1.5	
2	120	4.0	
3	180	4.0	
4	65	2.0	

	Famous_For
0	Iconic iron lattice tower, symbol of Paris
1	Bright lights, Broadway shows, New Year's Eve ...
2	World's most visited museum, home to Mona Lisa
3	Ancient defensive structure visible from space
4	White marble mausoleum, symbol of love

Data Info:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 30 entries, 0 to 29

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Place_Name	30 non-null	object
1	Country	30 non-null	object
2	City	30 non-null	object
3	Annual_Visitors_Millions	30 non-null	float64
4	Type	30 non-null	object
5	UNESCO_World_Heritage	30 non-null	object
6	Year_Built	30 non-null	object
7	Entry_Fee_USD	30 non-null	int64
8	Best_Visit_Month	30 non-null	object
9	Region	30 non-null	object
10	Tourism_Revenue_Million_USD	30 non-null	int64
11	Average_Visit_Duration_Hours	30 non-null	float64
12	Famous_For	30 non-null	object

```
dtypes: float64(2), int64(2), object(9)
memory usage: 3.2+ KB
```

Missing Values:

```
Place_Name      0
Country         0
City            0
Annual_Visitors_Millions  0
Type            0
UNESCO_World_Heritage  0
Year_Built      0
Entry_Fee_USD   0
Best_Visit_Month 0
Region          0
Tourism_Revenue_Million_USD 0
Average_Visit_Duration_Hours 0
Famous_For      0
dtype: int64
```

Summary Statistics:

	Place_Name	Country	City	Annual_Visitors_Millions \
count	30	30	30	30.000000
unique	30	14	23	NaN
top	Eiffel Tower	United States	New York City	NaN
freq	1	10	4	NaN
mean	NaN	NaN	NaN	11.331667
std	NaN	NaN	NaN	12.044247
min	NaN	NaN	NaN	1.500000
25%	NaN	NaN	NaN	4.775000
50%	NaN	NaN	NaN	7.575000
75%	NaN	NaN	NaN	12.250000
max	NaN	NaN	NaN	50.000000

	Type	UNESCO_World_Heritage	Year_Built	Entry_Fee_USD \
count	30	30	30	30.000000
unique	21	2	29	NaN
top	Historic Monument	Yes	1931	NaN
freq	3	17	2	NaN
mean	NaN	NaN	NaN	24.600000
std	NaN	NaN	NaN	28.572473
min	NaN	NaN	NaN	0.000000
25%	NaN	NaN	NaN	0.000000
50%	NaN	NaN	NaN	19.000000
75%	NaN	NaN	NaN	35.000000
max	NaN	NaN	NaN	109.000000

	Best_Visit_Month	Region	Tourism_Revenue_Million_USD \
count	30	30	30.000000

unique	15	10	NaN
top	Apr-May/Sept-Oct	North America	NaN
freq	5	10	NaN
mean	NaN	NaN	593.500000
std	NaN	NaN	1521.481894
min	NaN	NaN	25.000000
25%	NaN	NaN	61.250000
50%	NaN	NaN	87.500000
75%	NaN	NaN	167.500000
max	NaN	NaN	6800.000000

	Average_Visit_Duration_Hours	\
count	30.000000	
unique	NaN	
top	NaN	
freq	NaN	
mean	3.166667	
std	2.499425	
min	0.500000	
25%	1.500000	
50%	2.250000	
75%	4.000000	
max	10.000000	

	Famous_For
count	30
unique	30
top	Iconic iron lattice tower, symbol of Paris
freq	1
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

```
[11]: categorical_cols = ['Country', 'City', 'Type', 'UNESCO_World_Heritage',
    ↪ 'Best_Visit_Month', 'Region']

for col in categorical_cols:
    print(f"\nValue Counts for {col}:\n", df[col].value_counts())

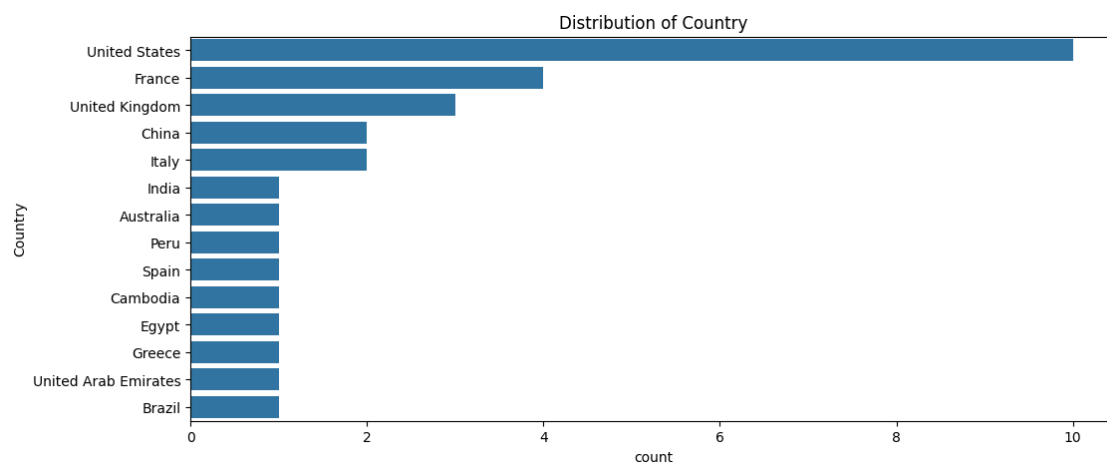
    # Plot
    plt.figure(figsize=(12,5))
    sns.countplot(data=df, y=col, order=df[col].value_counts().index)
    plt.title(f'Distribution of {col}')
```

```
plt.show()
```

Value Counts for Country:

Country	
United States	10
France	4
United Kingdom	3
China	2
Italy	2
India	1
Australia	1
Peru	1
Spain	1
Cambodia	1
Egypt	1
Greece	1
United Arab Emirates	1
Brazil	1

Name: count, dtype: int64

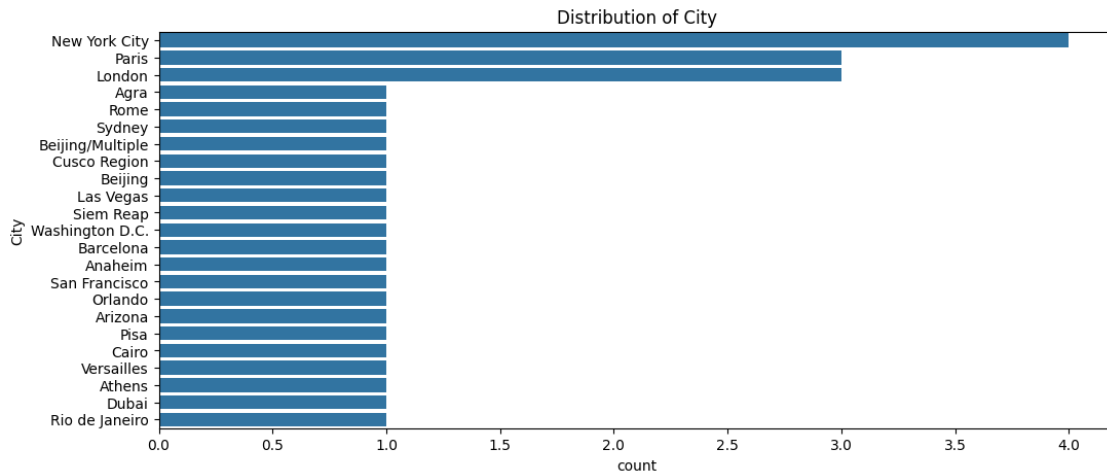


Value Counts for City:

City	
New York City	4
Paris	3
London	3
Agra	1
Rome	1
Sydney	1
Beijing/Multiple	1

Cusco Region	1
Beijing	1
Las Vegas	1
Siem Reap	1
Washington D.C.	1
Barcelona	1
Anaheim	1
San Francisco	1
Orlando	1
Arizona	1
Pisa	1
Cairo	1
Versailles	1
Athens	1
Dubai	1
Rio de Janeiro	1

Name: count, dtype: int64

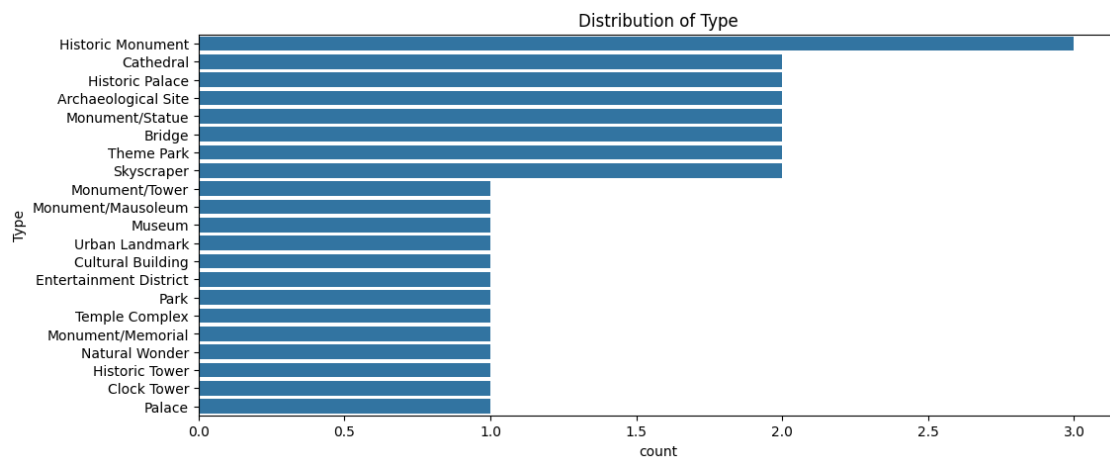


Value Counts for Type:

Type	count
Historic Monument	3
Cathedral	2
Historic Palace	2
Archaeological Site	2
Monument/Statue	2
Bridge	2
Theme Park	2
Skyscraper	2
Monument/Tower	1
Monument/Mausoleum	1

Museum	1
Urban Landmark	1
Cultural Building	1
Entertainment District	1
Park	1
Temple Complex	1
Monument/Memorial	1
Natural Wonder	1
Historic Tower	1
Clock Tower	1
Palace	1

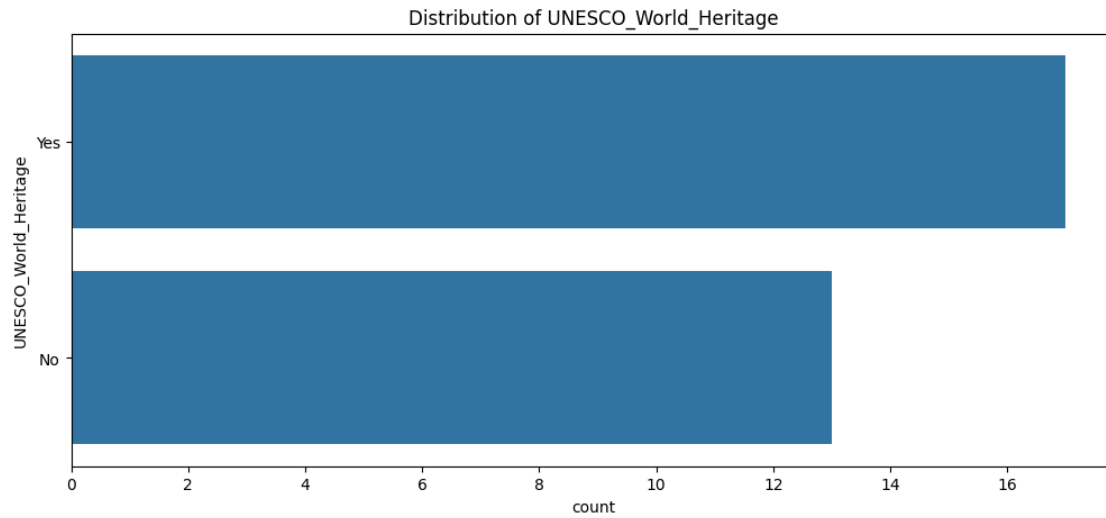
Name: count, dtype: int64



Value Counts for UNESCO_World_Heritage:

UNESCO_World_Heritage	
Yes	17
No	13

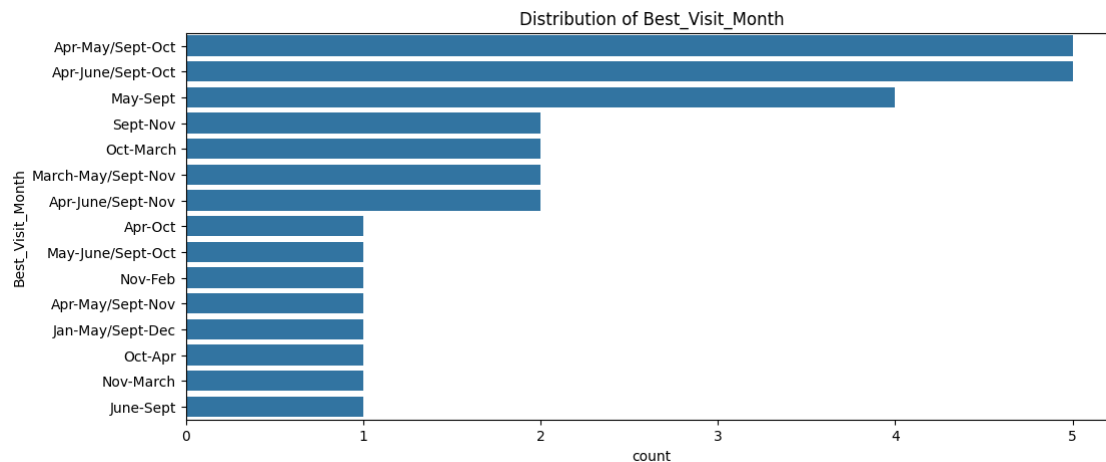
Name: count, dtype: int64



Value Counts for Best_Visit_Month:

Best_Visit_Month	
Apr-May/Sept-Oct	5
Apr-June/Sept-Oct	5
May-Sept	4
Sept-Nov	2
Oct-March	2
March-May/Sept-Nov	2
Apr-June/Sept-Nov	2
Apr-Oct	1
May-June/Sept-Oct	1
Nov-Feb	1
Apr-May/Sept-Nov	1
Jan-May/Sept-Dec	1
Oct-Apr	1
Nov-March	1
June-Sept	1

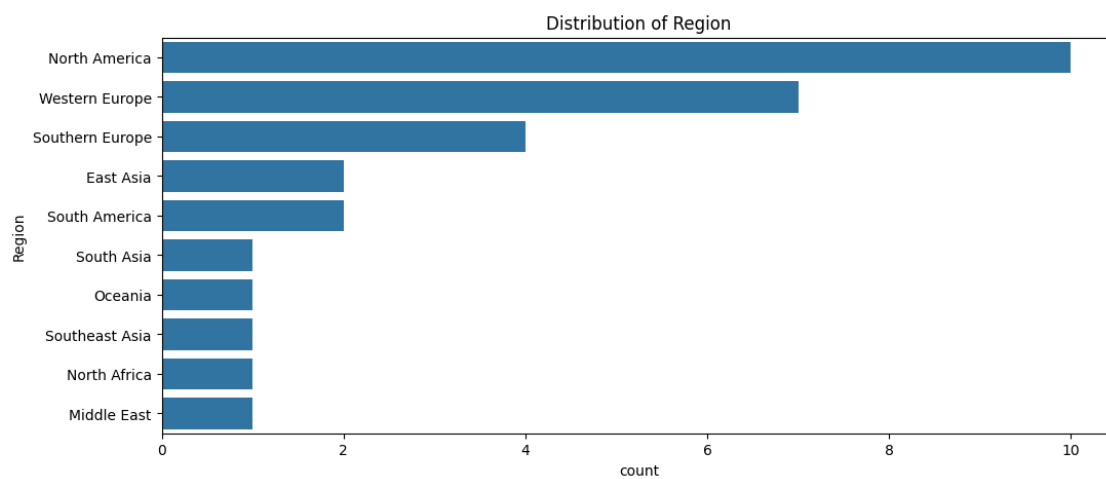
Name: count, dtype: int64



Value Counts for Region:

Region	count
North America	10
Western Europe	7
Southern Europe	4
East Asia	2
South America	2
South Asia	1
Oceania	1
Southeast Asia	1
North Africa	1
Middle East	1

Name: count, dtype: int64



Country Distribution

Insight: - The United States has the highest number of famous tourist places in the dataset.
- European countries like France and the UK also contribute significantly to global tourism.

City Distribution

Insight: - New York City appears most frequently, indicating it as a major global tourist hub.
- Cities like Paris and London also host multiple iconic attractions.

Attraction Type Distribution

Insight: - Historic monuments form the largest group, followed by museums, theme parks, and bridges. This shows that cultural and historical attractions dominate global tourism.

UNESCO World Heritage Distribution

Insight: - More than half of the tourist places are UNESCO World Heritage sites, highlighting the importance of cultural preservation in global tourism.

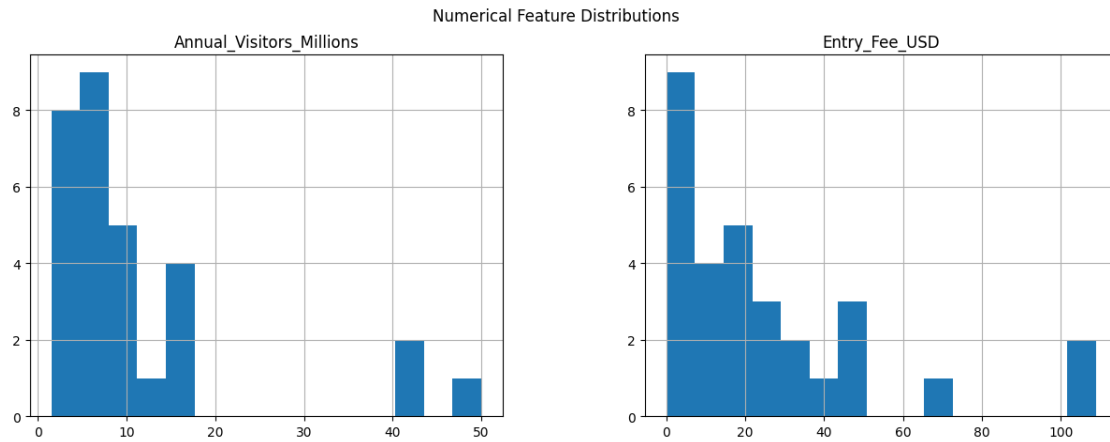
Region Distribution

Insight: - North America and Western Europe dominate the dataset.
- Other regions such as South Asia, Africa, and Southeast Asia have fewer famous places listed, indicating potential growth areas.

Numerical Feature Distributions

```
[12]: numerical_cols = ['Annual_Visitors_Millions', 'Year_Built', 'Entry_Fee_USD']

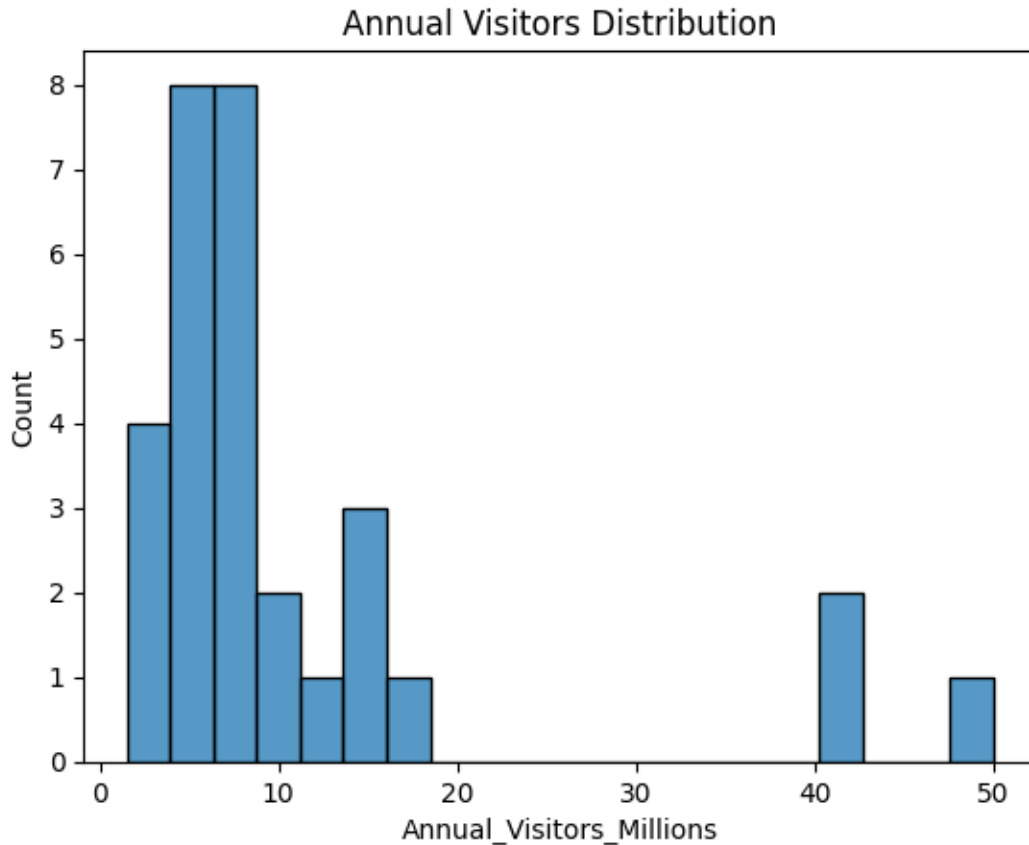
# Histograms
df[numerical_cols].hist(bins=15, figsize=(15,5))
plt.suptitle("Numerical Feature Distributions")
plt.show()
```



- Annual visitor counts are right-skewed, with a few very popular locations.
- Entry fees show high variation, from free attractions to premium-priced sites.

Annual Visitors Distribution

```
[13]: plt.figure()
sns.histplot(df['Annual_Visitors_Millions'], bins=20)
plt.title("Annual Visitors Distribution")
plt.show()
```



Most tourist places receive less than 15 million visitors annually, while a few major attractions act as outliers with extremely high footfall.

Total Annual Visitors by Attraction Type

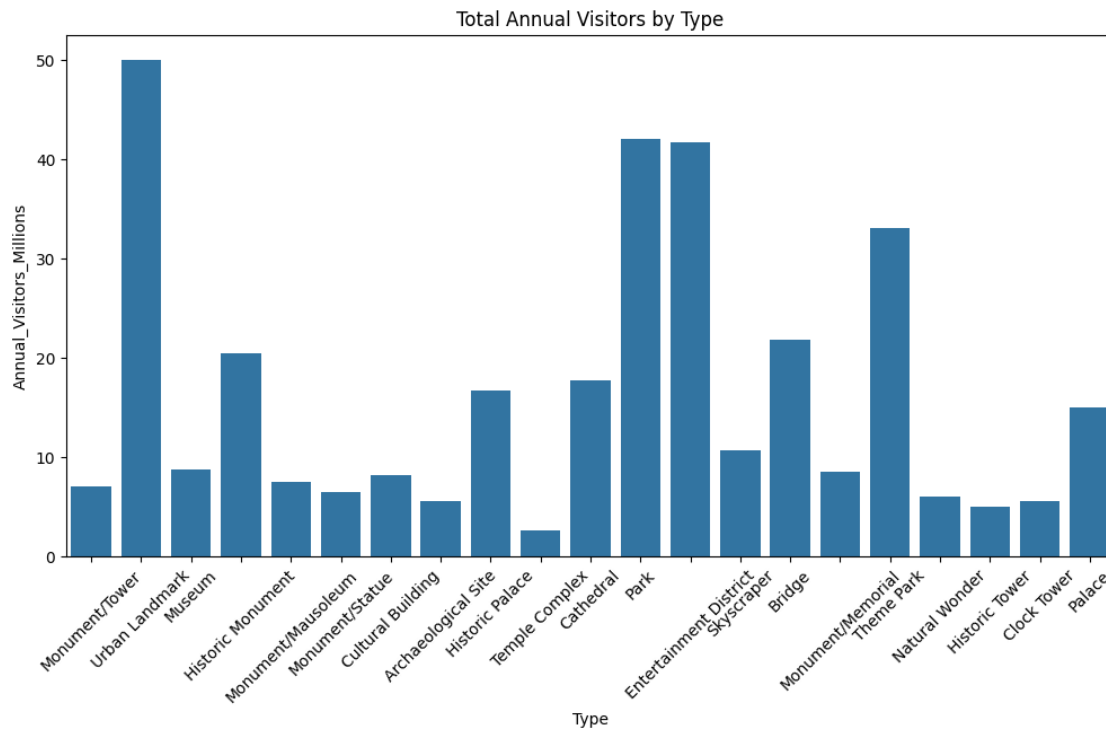
```
[16]: # 4. Relationship Between Features
# -----
# Annual visitors by Type
plt.figure(figsize=(12,6))
sns.barplot(data=df, x='Type', y='Annual_Visitors_Millions', ci=None,
            estimator=sum)
plt.title("Total Annual Visitors by Type")
plt.xticks(rotation=45)
plt.show()
```

C:\Users\Riya\AppData\Local\Temp\ipykernel_19792\2098112134.py:5: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

```
sns.barplot(data=df, x='Type', y='Annual_Visitors_Millions', ci=None,
```

```
estimator=sum)
```

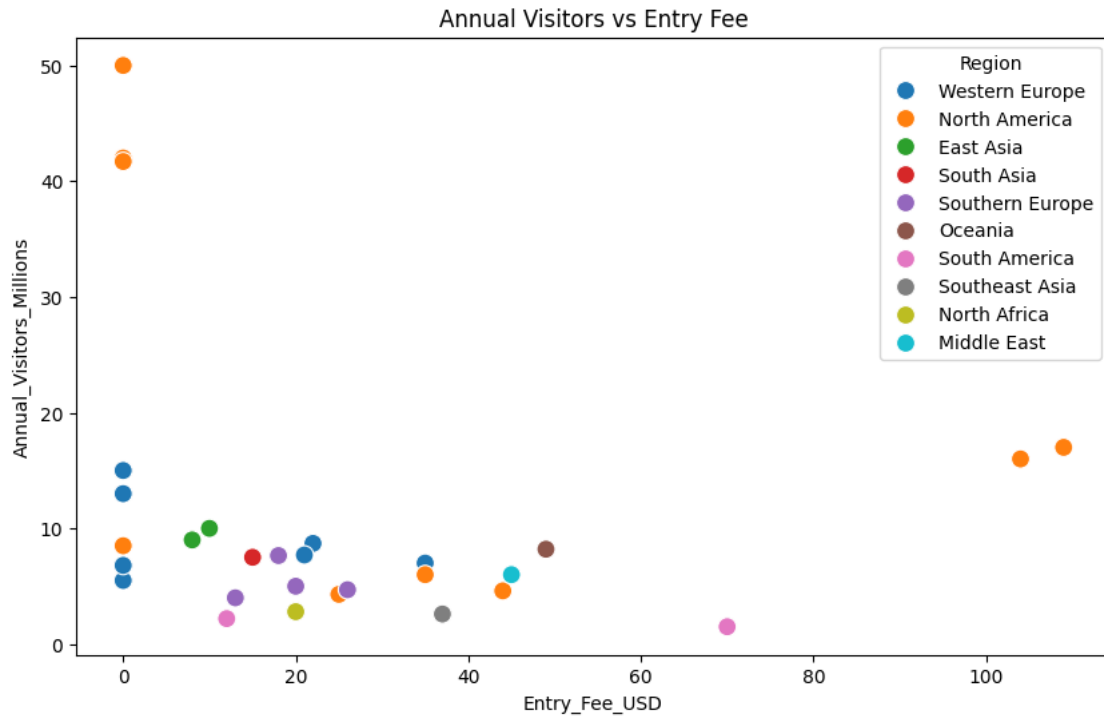


- Urban landmarks and entertainment-focused attractions attract the highest total visitors.

- Traditional monuments and heritage sites show stable but comparatively lower visitor volumes.

Visitors vs Entry Fee

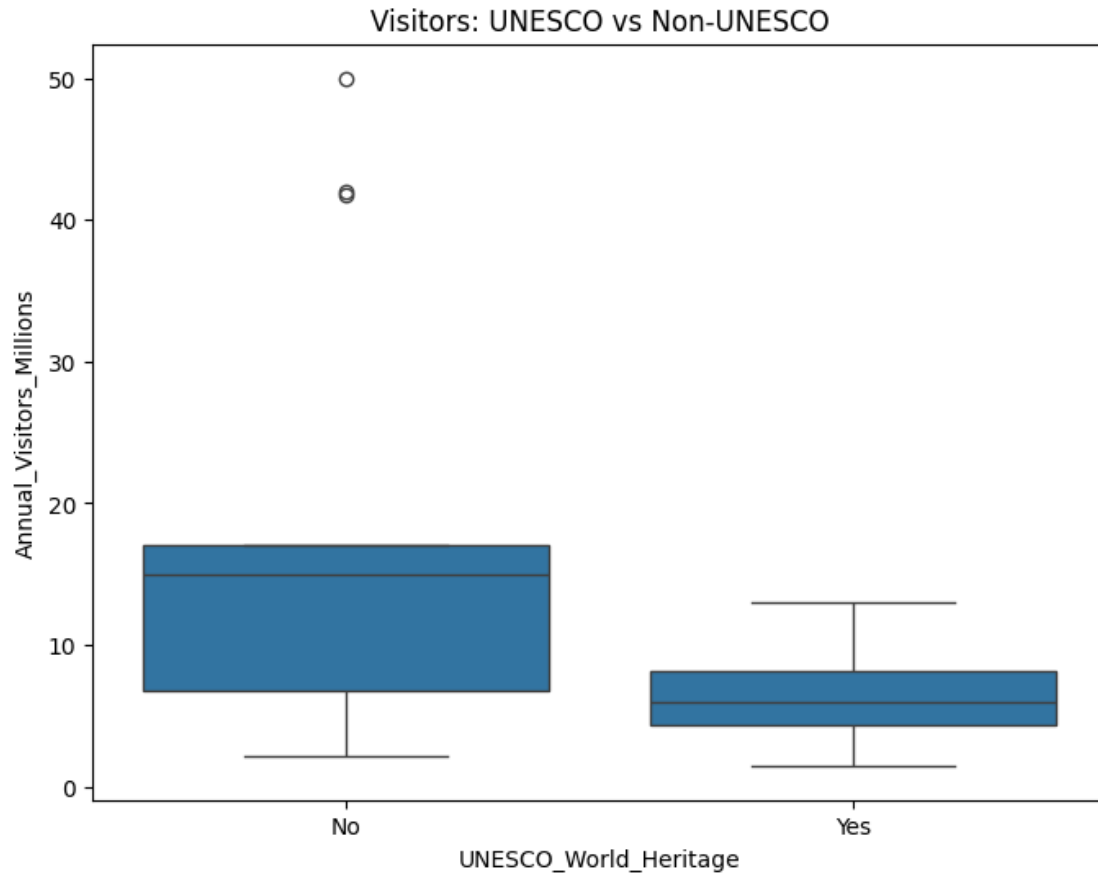
```
[24]: # Visitors vs Entry Fee
plt.figure(figsize=(10,6))
sns.scatterplot(data=df, x='Entry_Fee_USD', y='Annual_Visitors_Millions',
               hue='Region', s=100)
plt.title("Annual Visitors vs Entry Fee")
plt.show()
```



There is no strong negative relationship between entry fee and visitor count. Many high-fee attractions continue to attract a large number of tourists.

UNESCO vs Non-UNESCO Visitors

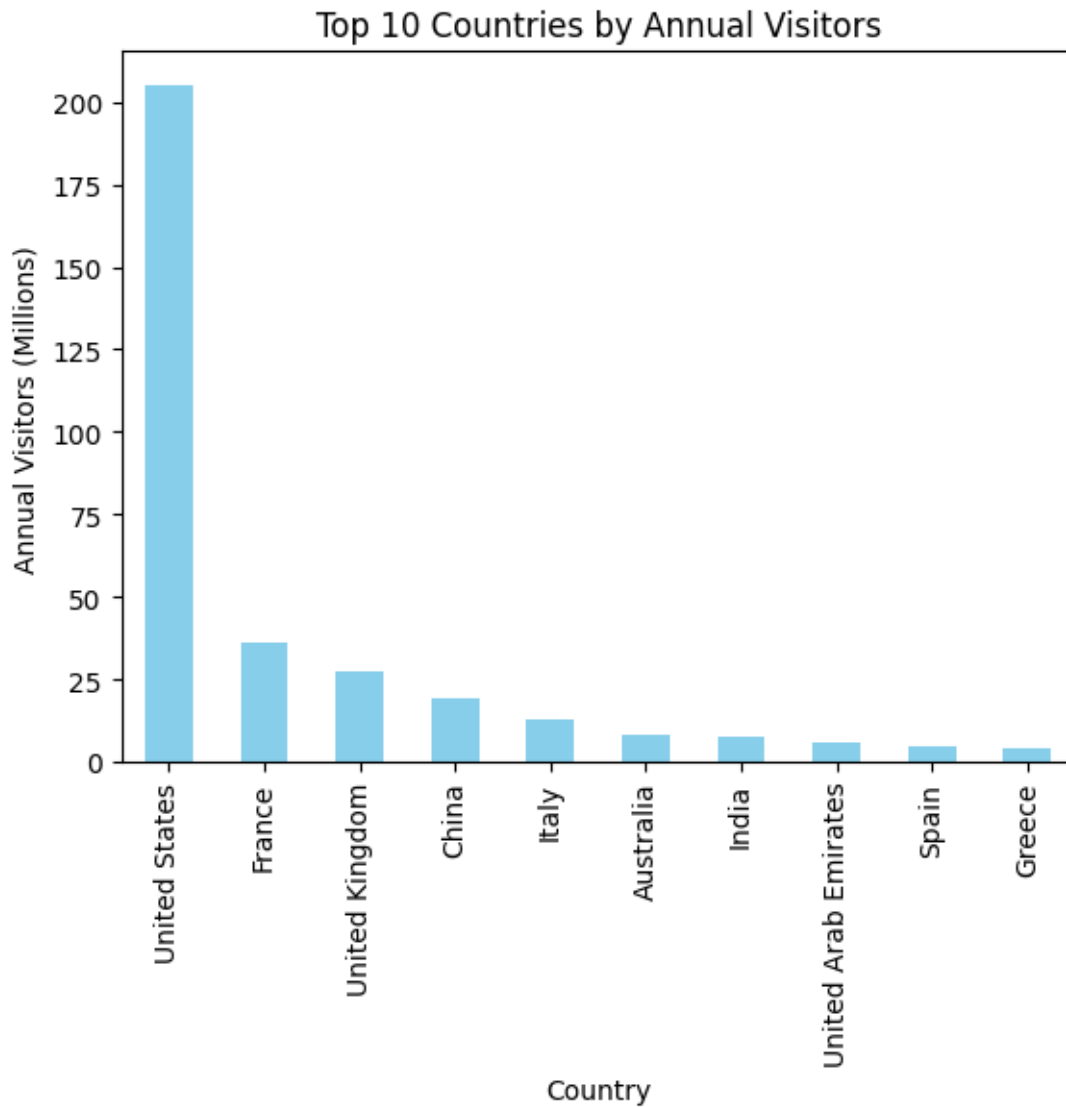
```
[25]: # UNESCO vs Non-UNESCO Sites Visitors
plt.figure(figsize=(8,6))
sns.boxplot(data=df, x='UNESCO_World_Heritage', y='Annual_Visitors_Millions')
plt.title("Visitors: UNESCO vs Non-UNESCO")
plt.show()
```

Non-UNESCO sites show higher median visitor counts, mainly due to popular urban landmarks, while UNESCO sites show more consistent visitor numbers.

Top Countries by Annual Visitors

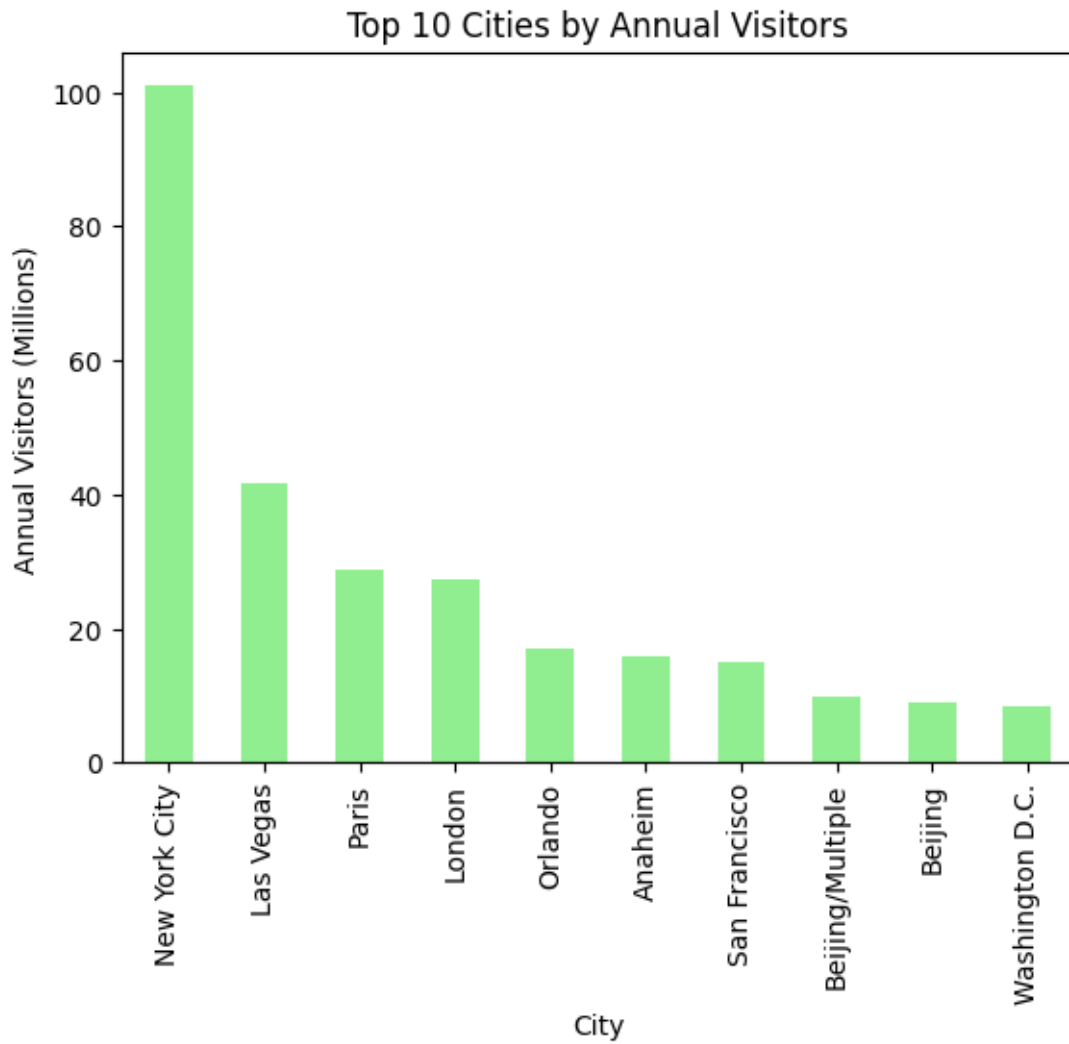
```
[27]: # 5. Top Countries and Cities by Visitors
# -----
top_countries = df.groupby('Country')['Annual_Visitors_Millions'].sum().
    ↪sort_values(ascending=False).head(10)
top_countries.plot(kind='bar', color='skyblue', title="Top 10 Countries by
    ↪Annual Visitors")
plt.ylabel("Annual Visitors (Millions)")
plt.show()
```



The United States leads by a large margin in total annual visitors, followed by France and the United Kingdom.

Top Cities by Annual Visitors

```
[28]: top_cities = df.groupby('City')['Annual_Visitors_Millions'].sum().  
      ↪sort_values(ascending=False).head(10)  
top_cities.plot(kind='bar', color='lightgreen', title="Top 10 Cities by Annual_  
      ↪Visitors")  
plt.ylabel("Annual Visitors (Millions)")  
plt.show()
```



- New York City ranks highest due to multiple high-traffic attractions.

- Other major cities like Las Vegas, Paris, and London also attract significant tourist volumes.